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Q1) Explain learning factors with example
Q2) Explain learning Rules with an example

Q1) ⇒ Learning Factors
i) Initial weights
ii) Cummulative vs incremented updation
iii) learning Rate
iv) learning Constant & momentum
v) steepness of neuron

i) Initial ~~Rate~~ weight:-
Initial weight is the Input weight of the neuron
with is must be initialized randomly
when Initial weight is initialized to zero
it gives the same output it's not optimal

example) there are 5 inputs of neuron
x_1, x_2, x_3, x_4, x_5 and which initially declared
to some random values which is anything.

ii) Cummulative vs Incremetation updation of weights

cummulative weight updation (Batch updation)
- the weights are updated after the whole iteration of training model on the neuron
- after all neuron activation weights are cummulatively updated

Incrementation updation of weight (online updation)
- The weights are updated after the each activation of the neuron
- emidiately after the the ~~ut~~ activation the each neuron weight is updated

iii) Steepness of neuron
Steepness of neuron is the seriousness of the neuron
When
if activation is too steep → large gradient → unstable.
if activation is flat → slow learning

iv) learning Constant (η)
learning constant is used for managed neuron converance

$W(i) = \eta \, x_i \, y$

where η = learning constant
x_i = input weights
y = output

when η is too large output is overfitted and undesired
, η is too small the output is predected

v) momentum (α)
momentum is for calculating the converance of the neuron
it measures the oscilation
and the speed of the neuron
3 rapidity of neuron is calculated by the momentum

Q.2) ⇒ Learning Rules

i) Hebbien Learning Rule
ii) Peceptron Learning Rule
iii) Corelation Learning Rule
iv) delta Leaning Rule
v) Widrof - Hoff - Learning Rule
vi) Winner Take all Learning Rule

i) Hebbian Learning Rule
- It is a Initial Learning Rule
- it is unsupervised learning Rule in which the no targeted output is there
- no error correction or updation

The Principle of the Rule
:- when All neurons fire together, that are wire together

$$W(i) = \eta x_i y$$

In this rule the associative learning are used internally the all neurons are Interrelated

example) Feature learning, clustring

iii) delta learning Rule
(gradient decent)

it is also supeervised learning Rule
it is primarily build for the
· ADALINE and multilayer network

when the Mean square error is minimized
and updated using the error function

the gradient is the calculas function
which is being used the cal loss

$w(i) = \eta \frac{\delta L}{\delta W}$

$error = \frac{\eta (t-y)^2}{2}$

example) classification & regression
problem

ii) Perceptron Learning Rule

it is a supervised learning Rule
it is used for the single Layer perceptron

in this learning Rule the error updation is there

$$W(i) = \eta (t-y) x_i$$

where t = targeted output
y = actual output
x_i = input

ex) binary classification

iii) corelation learning Rule

this is the associative learning Rule
when the targeted output is related to input and actual output

it is also supervised learning
where the input, targeted output are preknown

$$W(i) = \eta x_i (t-y)$$

error minimization is used

v) widrof-Hoff learning Rule

it is developed by two scientiest widrof and Hoff

it is also supervised learning Rule the targeted output is pre known

it is special form of the delta learning Rule
it is used for the ADALINE network only
in this Rule error minimization is used

$$w(i) = \eta (t-y) x_i$$

example) ⇒ classification, analysis

vi) winner-Take all

it is unsupervised learning rule

in this Rule the activation which has highest it only update the weight

all rights is gives to winner the activation which has highest weight and activation

4 example) feature selection